

Mid term review

Exercise 1 : TCP/ UDP

Suppose you are designing the communication part of a multi-functional teleconferencing app. Your teleconferencing app allows voice talk and text-based messaging among the users. The table below shows the QoS (quality of service) the general users demand for the two functions.

	Loss	Bandwidth	Time Sensitive
=====			
Messaging	no loss	elastic	no
Talk	loss tolerant	audio: 5kbps	yes

1. Which transport layer services, TCP or UDP, will you choose to transfer the text messages and why?
2. Which transport layer services, TCP or UDP, will you choose to transfer the talk voices and why?

Exercise 2 :

1. What information is used by a process running on one host to identify a process running on another host?
2. Suppose you wanted to do a transaction from a remote client to a server as fast as possible. Would you use UDP or TCP? Explain why.
3. Why do HTTP, SMTP, and POP3 run on top of TCP rather than on UDP?
4. Suppose a user has two browser applications active at the same time, and suppose that the two applications are accessing the same server to retrieve HTTP documents at the same time. How does the server tell the difference between the two applications?
5. Give two different examples of application-layer protocols that rely on caching to improve the user-perceived performance of the network application. Make sure to indicate why caching is used, where the caches are located, and how they work.
6. Suppose that Alice wants to send an email message to Bob. This will involve four entities: Alice's mail client (for email composition and sending), Alice's outgoing mail server, Bob's incoming mail server, and Bob's mail client (for email retrieval and viewing). Between which of these four entities does the SMTP protocol operate? What about the IMAP protocol?

7. What is the purpose of the HTTP “COOKIE:” field? Are the values in the HTTP message’s cookie field stored at the client or server or both? Explain briefly.
8. Define each term, and clarify the key difference(s) between them: “client-server” and “peer-to-peer”
9. While watching a recorded lecture on Canvas, you open a second browser tab to scroll through your favorite cat-related blog. You can assume that these are the only applications connecting to the Internet on your machine, that each tab (Canvas + cat blog) opens TCP sockets, and that your machine has one IP address. Given this, when a TCP segment arrives, how does the OS figure out which application to deliver it to? (In other words, what does your OS do to make sure that a cat doesn’t end up in your lecture video?)

Exercise 3 :

Suppose that a lunar rover robot on the Moon takes a 2 MB “selfie” photo and transmits it home to its parent robot on Earth. The transmission uses an error-free direct link with a data transmission rate of $R = 4$ Megabits per second (Mbps).

1. Calculate the transmission time for this file, which has size L (in bits). Recall that $1 \text{ Mbps} = 10^6$ bits per second. Show your work.
2. Assuming that the Moon is approximately 385,000 kilometers from the Earth, at what time would the very first bit of the photo arrive? Recall that the speed of light is approximately 3×10^8 meters per second. Show your work.